Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading dBc off a Tracking-tab trace.** A cadet walks an HB100 around the PHASER on a 1 m arc with the beam held at broadside and records the Signal vs Time trace. The three lobe peaks read −18.4 dBFS (main lobe), −31.2 dBFS (left first sidelobe), and −29.9 dBFS (right first sidelobe).

   (a) Convert both sidelobe readings to dBc.

   $L_{\text{left}} = \boxed{}$

   $L_{\text{right}} = \boxed{}$

   (b) The cadet's partner raises **Rx Gain** by 6 dB and repeats the walk. State what happens to the three dBFS readings and what happens to the two dBc values, and explain why the two answers differ.

   (c) A uniform 8-element array has its first sidelobes at −13 dBc. Comment on the pair measured above: is this array uniformly illuminated, and what accounts for the 1.3 dB spread between the two sides?

2. **Why the time axis is not an angle axis.** The same trace runs from $t = 0.00$ s, when the source was at $-90°$, to $t = 7.40$ s, when it reached $+90°$. The main-lobe peak lands at $t = 3.10$ s. The beam was commanded to broadside.

   (a) Explain why the lobe *amplitudes* on this trace are usable while the lobe *angles* are not.

   (b) Assume a constant walking speed and convert the main-lobe time to an angle. Compare it against the commanded beam angle and state the error.

   $\theta =$ $\boxed{}$

   error = $\boxed{}$

   (c) The true half-power beamwidth is 13.1°. If the hand moves as much as 30% faster or slower than its average while crossing the main lobe, give the range of beamwidths a constant-speed conversion would report.

   range = $\boxed{}$

   (d) Name the piece of range hardware that removes this error and state what it records that the hand walk does not.

3. **Scan loss and sidelobe asymmetry.** The projected-aperture rule puts the peak power gain of a steered array at $\cos\theta_0$ relative to broadside.

   (a) Compute the predicted peak drop at steer angles of 30°, 45°, and 60°.

   $30^\circ =$ $\boxed{}$

   $45^\circ =$ $\boxed{}$

   $60^\circ =$ $\boxed{}$

   (b) A broadside walk peaks at −18.4 dBFS and a 30° walk peaks at −19.3 dBFS. Give the measured scan loss, and state whether one pair of walks resolves it given ±0.5 dB of run-to-run scatter.

   $\Delta =$ $\boxed{}$

   (c) The measured 30° trace shows first sidelobes of −12.2 dBc on the low-angle side and −15.2 dBc on the high-angle side. Explain why they are unequal, given that the array factor alone would make them equal.

   (d) Real patch elements follow a power pattern nearer $\cos^{1.4}\theta$ than $\cos\theta$. Recompute the 60° scan loss and state how much worse it is than the ideal-element bound.

   loss = $\boxed{}$

4. **How far out is far enough.** The PHASER's 8 elements are spaced 14 mm apart, so the aperture spans $D = 98$ mm between the outer element centers. The HB100 runs at 10.525 GHz.

   (a) Find the far-field distance for this aperture and state whether the 1 m arc used in lab is far enough.

   $r =$ $\boxed{}$

   (b) A Ku-band panel with a 0.200 m aperture is to be measured at 15.0 GHz. Find its far-field distance.

   $r =$ $\boxed{}$

   (c) The Ku-band panel is measured on the same 1.0 m classroom arc anyway. Find the phase error across the aperture edge and compare it against the 22.5° tolerance behind the far-field criterion.

   $\Delta\phi =$ $\boxed{}$

   (d) Describe what a 90° edge phase error does to the measured pattern, and say which pattern feature it corrupts worst.

5. **Classroom against chamber.** The lab trace carries about ± 1 dB of ripple from room reflections adding to the direct path.

   (a) List which of the four hand-measurement artifacts from the lab an anechoic chamber removes, and which one it does not remove by itself.

   (b) A single reflected path of relative amplitude ρ adds to a direct path of amplitude 1. Find the ρ that produces a $+1$ dB peak in the received amplitude, and express it in dB.

   $\rho = \boxed{}$

   (c) The main-lobe peak and the sidelobe peak each carry their own ± 1 dB of ripple. Give the worst-case uncertainty on a sidelobe level reported in dBc, and give the range a true -13 dBc sidelobe could read.

   uncertainty = $\boxed{}$

   range = $\boxed{}$

   (d) A customer requires written proof that this array's sidelobes are below -25 dBc. State whether the classroom measurement can support that claim, and give the two reasons that decide it.

Documentation: ______________________________